ENV 710 Initial Figures

S. Slay, E. Miles-Flynn, & Y.E. Su

2026-03-19

```r
# Load the packages
library(here)
library(tidyverse)
library(moments)
library(dplyr)
library(viridis)
library(ggplot2)
library(ggpath)
library(gt)
library(paletteer)
library(grid)
library(gridExtra)
library(scales)
library(kableExtra)
library(knitr)
library(ggcorrplot)
library(readr)
```

```r
# Set global options for R chunks
knitr::opts_chunk$set(echo = TRUE,
                      message = FALSE,
                      warning = FALSE)
```

Project files

The project is stored in the following Git repository: https://github.com/emilesflynn/ENV_710_Final_Project/

The raw data was downloaded from the Durham water quality web portal, which is managed by the City of Durham: http://www.durhamwaterquality.org/

Data Wrangling

The raw data frame presents each measurement in a separate row. The code below merge the measurements conducted at the same location on the same day into one observation.

```r
# Set your ggplot theme
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "right")
```

```r
theme_set(mytheme)

# Load your datasets
durham_data_raw <- read_csv(here("Data/SlaySMilesFlynnESuEData_Raw_2025.csv"),
    col_types = cols(Date = col_date(format = "%Y/%m/%d")))
# View(durham_data_raw)

# Pivot table and merge the measurements
df_wide <- durham_data_raw %>%
  pivot_wider(
    id_cols = c(`Station Name`, Date),
    names_from = Parameter,
    values_from = Value,
    values_fn = mean
  )

write.csv(df_wide, file = here("Data/SlaySMilesFlynnESuEData_Cleaned.csv"), row.names = FALSE)
```

After obtaining the clean data frame, we then went on to add more explanatory or informational columns to help with further analyses. We isolated the river name and the sampling locations' distances to the river mouth, and add the year and month columns to help visualize the trends on the temporal scale.

```r
# Load the datasets
here()
```

```
## [1] "/Users/ericarquette/ENV710Project"
```

```r
data_clean <- read_csv(here("Initial Figures/SlaySMilesFlynnESuEData_Cleaned_2025.csv"))

data_clean <- data_clean %>%
  mutate(
    #Extract the river name abbreviation from the station name
    River_abbr = str_extract(`Station Name`, "^[A-Z]{2}"),
    #Extract the distance to stream mouth
    Distance_miles = as.numeric(str_extract(`Station Name`, "[0-9.]+"))
  )

# Find rows where the distance extraction failed
problem_rows <- data_clean%>%
  mutate(temp_dist = str_extract(`Station Name`, "[0-9.]+")) %>%
  filter(is.na(as.numeric(temp_dist)) & !is.na(`Station Name`))

# Show the unique station names that failed
unique(problem_rows$`Station Name`)
```

```
## character(0)
```

```r
# Replace the abbreviations with your actual names
river_lookup <- data.frame(
  River_abbr = c("EL", "EN", "NH", "TF", "CC", "NE", "LC", "LL", "LR", "PN", "SI"),
  Watershed = c("Ellerbe Creek", "Eno River", "New Hope Creek", "Third Fork Creek",
                "Crooked Creek", "Northeast Creek", "Lick Creek", "Little Lick Creek",
```

```r
                        "Little River", "Panther Creek", "Stirrup Iron Creek")
)

# Merge the data frames
data_clean_final <- data_clean %>%
  left_join(river_lookup, by = "River_abbr")

# Add month and year columns
data_clean_final$Date <- ymd(data_clean_final$Date)

data_clean_final <- data_clean_final %>%
  mutate(Month = factor(month(Date),
                        levels = 1:12,
                        labels = month.abb,
                        ordered = TRUE),
         Year = year(Date),
         Storm_status = case_when(Date <= as.Date("2025-07-03") ~ "Before",
                                  Date >= as.Date("2025-07-09") ~ "After"),
         Basin = case_when(
           River_abbr %in% c("EL", "EN", "LC", "LL", "LR", "PN", "SI") ~ "Neuse (North)",
           River_abbr %in% c("CC", "NE", "NH",  "TF") ~ "Cape Fear (South)"))

# Reorder columns
data_clean_final_2025 <- data_clean_final %>%
  select(`Station Name`, Watershed, River_abbr, Distance_miles, Basin,
         Date, Month, Year, Storm_status, everything())

write.csv(data_clean_final_2025, file = here("Data/SlaySMilesFlynnESuEData_Final_2025.csv"),
          row.names = FALSE)

# Combined the dataset
# Change the year above to 2024 to obtain the 2024 data frame
data_2024 <- read.csv(here("Data/SlaySMilesFlynnESuEData_Final_2024.csv"))
data_2025 <- read.csv(here("Data/SlaySMilesFlynnESuEData_Final_2025.csv"))
data_combined_2425 <- bind_rows(data_2024, data_2025) %>%
  mutate(Date = as.Date(Date),
         Year = as.factor(year(Date)),
         # Extract the numeric month (1-12) for the x-axis calculation
         Month_Num = month(Date))
```

Intial figures

Trends over the time

This figure shows how the concentration of Dissolved Oxygen (mg/L) changed over time before and after the storm. The vertical line indicates when the Tropical Storm Chantal occurred on July 6, 2025.

```r
# Comparing between Northern and Southern Basins
do_plot_2025 <- ggplot(data_clean_final_2025, aes(x = Date, y = `Dissolved Oxygen`,
                                                  color = Basin)) +
  # Add points and lines (grouped by Basin to show trends)
  geom_point(alpha = 0.5, na.rm = TRUE) +
```

```r
  geom_smooth(se = FALSE, method = "loess", na.rm = TRUE) +

  # Add the NC DEQ Water Quality Standard line (5.0 mg/L)
  geom_hline(yintercept = 5.0, linetype = "dashed", color = "darkorange") +
  annotate("text", x = min(data_clean_final$Date), y = 5.5,
           label = "NC Standard (5.0 mg/L)", color = "red", hjust = 0) +

  # Add a vertical line to represent TS Chantal
  geom_vline(xintercept = as.Date("2025-07-06"),
             linetype = "dashed",
             color = "darkred",
             linewidth = 0.5) +
  # Add a text label next to the line
  annotate("text",
           x = as.Date("2025-07-06"),
           y = 11,
           label = "TS Chantal (Jul 6)",
           color = "darkred",
           angle = 90,
           vjust = -0.5,
           fontface = "bold")+

  # Extend the x-axis to show the full labels
  scale_x_date(
    limits = c(as.Date("2025-01-01"), as.Date("2026-01-01")),
    date_labels = "%b %Y",
    date_breaks = "4 months")+

  # Formatting
  scale_color_manual(values = c("Neuse (North)" = "#2c7bb6",
                                "Cape Fear (South)" = "#d7191c")) +
  labs(title = "Dissolved Oxygen Levels in Durham Streams (2025)",
       subtitle = "Comparing Northern (Neuse) and Southern (Cape Fear) Watersheds",
       x = "Sampling Date",
       y = "Dissolved Oxygen (mg/L)") +
  theme_minimal() +
  theme(legend.position = "bottom")
do_plot_2025
```

Scatterplot for linear regression models

The figure below shows the relationship between Total Phosphorus (TP), which is a variable showing aquatic nutrient pollution level, and Dissolved Oxygen (DO), which is a strong aquatic health indicator. The linear regression model finds a strong relationship (p-$value < 0.05$).

```r
regression_plot <- data_clean_final_2025  %>%
  # This removes all rows where the x-variable is greater than 0.65
  filter(`Total Phosphorus` <= 0.65) %>%
  ggplot(aes(x = `Total Phosphorus`, y = `Dissolved Oxygen`,
                                        color = Basin)) +
  # Use points with transparency to handle overlapping data
  geom_point(alpha = 0.5, size = 2, na.rm = TRUE) +
```

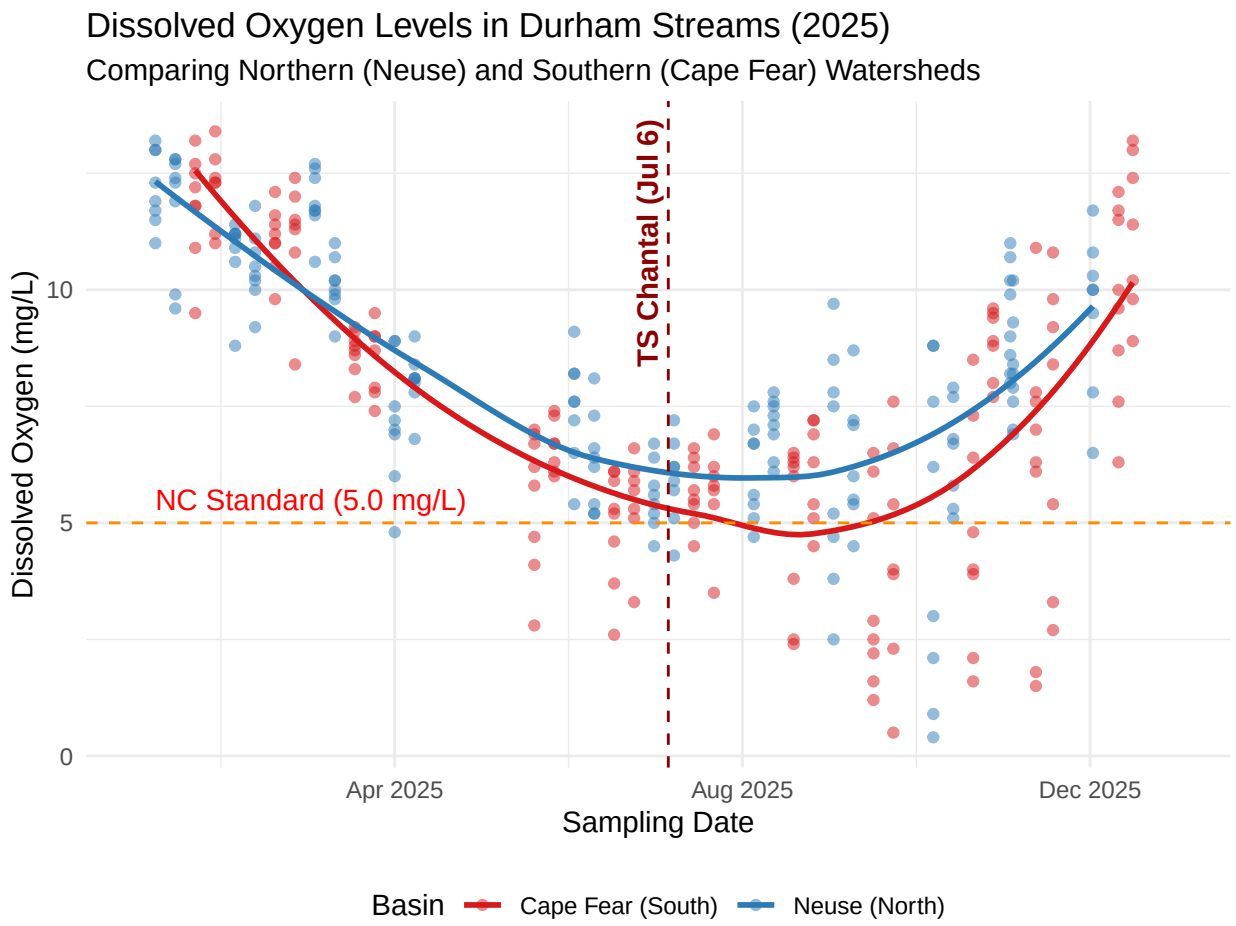

Figure 1: 2025 Durham Stream Dissolved Oxygen Levels Streams by Basin.

```r
  # Add the Linear Regression Model (method = "lm")
  # This adds a line of best fit for each Basin
  geom_smooth(method = "lm", se = TRUE, fill = "lightgray", na.rm = TRUE) +

  # Formatting
  scale_color_brewer(palette = "Set1") +
  # facet_wrap(~Basin) +
  labs(title = "Relationship Between Total Phosphorus and Dissolved Oxygen",
       subtitle = "Linear regression of Total Phosphorus vs. Dissolved Oxygen by Basin",
       x = "Total Phosphorus (mg/L)",
       y = "Dissolved Oxygen (mg/L)",
       color = "Basin",
       caption = "Three outliers (TP > 0.65 mg/L) removed as high-leverage observations") +
  theme_minimal() +
  theme(legend.position = c(0.8, 0.15))

regression_plot
```

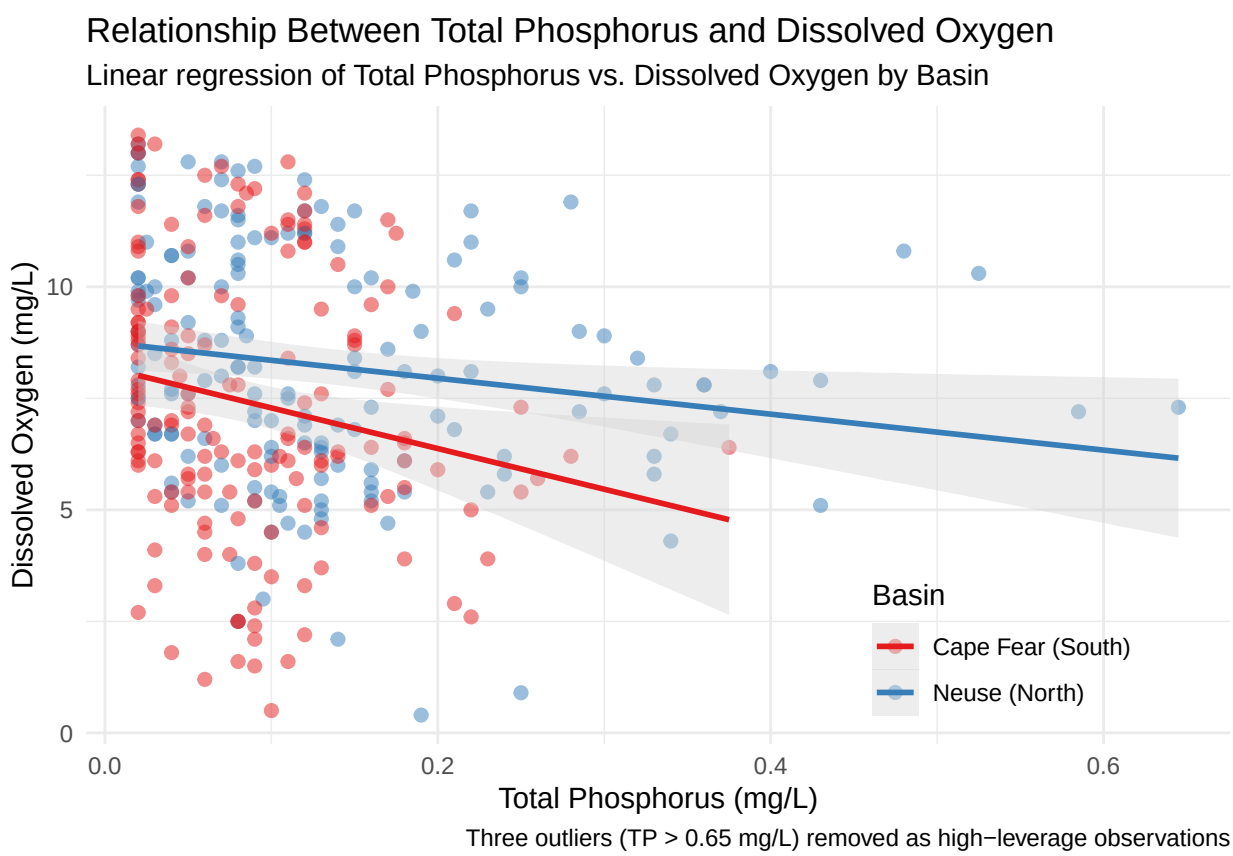

Figure 2: Linear regression of Total Phosphorus v. Dissolved Oxygen by Basin.

The figures below are exploratory.